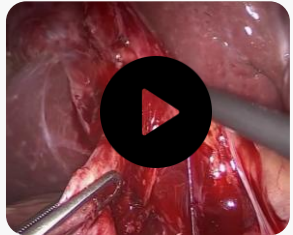


Input

Surgical Videos



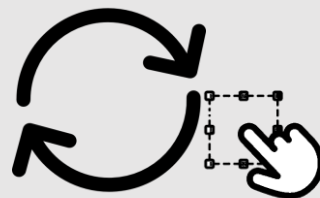
+

Meta Data

e.g. Instrument
Localization and/or
Action Triplet
Information

SurgSTU-Pipeline

Metadata Extension&Structuring

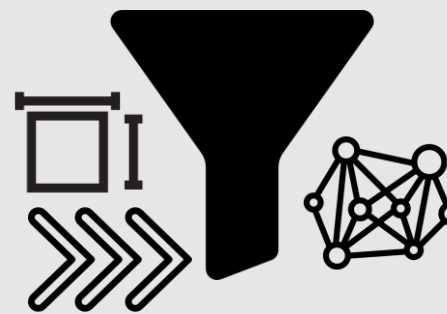


Extending videos with
surg.instrument
localization
information

$(fn, c_i, bbox, c_v, c_t)$
 $(fn, c_i, bbox, c_v, c_t)$
 $(fn, c_i, bbox, c_v, c_t)$

Metadata parsing and
consolidation into
structured Event Tuple
format

Spatial-Temporal QA Generation



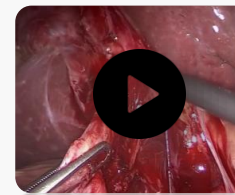
Spatial/Temporal/Sema
ntic **Continuity Filtering**



Completing
predefined **QA**
templates

Output

Spatial-Temporal
Vision-Language
Dataset



Q: What does the surg.
instrument [..]
A: After idling, the
scissors [..]